1828

An illustration from Audubon's book, *The Birds of America*.

Originally set up as a place for scientific study, the Gardens of the London Zoological Society (London Zoo) was eventually opened to the public in 1847.

(1802–76) **discovered bromine in seawater**. And in Tours, French doctor **Pierre Bretonneau** (1778–1862) identified **diphtheria**.

In 1827, English chemist **William Prout** (1785–1850) **classified food into the three main divisions** known today: carbohydrates, fats, and proteins.

Scottish naturalist **Robert Brown** (1773–1858) observed the phenomenon now known as **Brownian motion** (see panel, opposite). French–American naturalist **John James Audubon** (1785–1851) sold the first prints of his book, *The Birds of America, a 13-year project that culminated in 1838 with a total of 435 plates*.

> THE **OBJECTS** APPEAR WITH **ASTONISHING SHARPNESS…** DOWN TO THE SMALLEST DETAILS… THE **EFFECT** IS DOWNRIGHT **MAGICAL.**
>
> **Joseph Nicéphore Niépce, French inventor,** describing an earlier photographic experiment to his brother, September 16, 1824

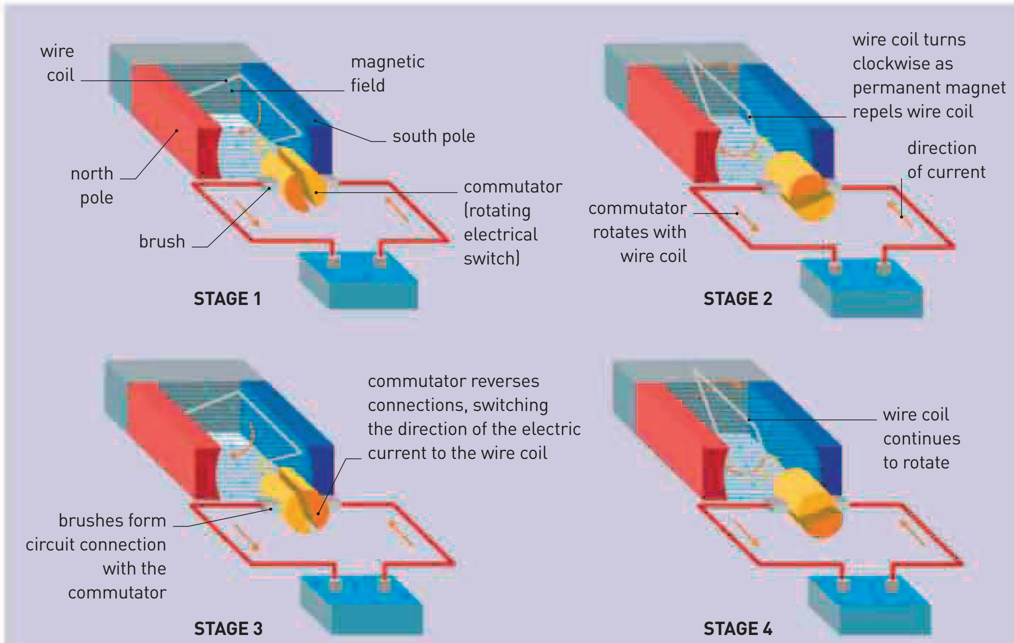
AS THE WESTERN WORLD BECAME INCREASINGLY URBANIZED, interest turned toward the natural world and **botanic gardens and zoos** were opened to display exotic plants and animals. The first zoo for scientific study was London Zoo in England, which opened on April 27, 1828.

The **foundations of embryology** (the science of embryo development) were laid by Estonian naturalist **Karl Ernst von Baer** (1792–1876), who showed that different animal species could appear similar at early stages of development.

English fossil collector **Mary Anning** (see 1810–11) had made a number of important prehistoric discoveries on the English coast, but in 1828 she found the **fossil of a pterosaur**, a huge flying reptile. It was only the third such fossil to be found and the first to be identified. Anning's pterosaur was named *Dimorphodon* by paleontologist Richard Owen (1804–92) in 1859.

In Berlin, German chemist **Friedrich Wöhler** (1800–82) **pioneered organic chemistry**—the chemistry of living things—with his discovery of what is now called **Wöhler synthesis**. This is a chemical reaction that produces the organic chemical urea. Its discovery showed that organic chemicals were not only made by living things, as Wöhler's former tutor Swedish chemist **Jön Jakob Berzelius** (1779–1848) had insisted. Berzelius also made a discovery in 1828 when he **isolated the radioactive element thorium**, a dense metal found in a black mineral discovered by the Norweigian geologist Morten Esmark (1801–82).

The world's **first electric motor** was made in Budapest by Hungarian inventor and Benedictine monk **Ányos Jedlik** (1800–95), and in Nottingham, England, self-taught mathematician **George Green** (1793–1841) published an essay in which he outlined the first **mathematical theory of electricity and magnetism**. This was later built on by James Clerk Maxwell (see 1861–64).



ELECTRIC MOTORS

Hungarian inventor Ányos Jedlik showed how an electric motor could be driven by the repulsion between the poles of a permanent magnet and an electromagnetic coil. A half turn moves the like poles of the coil away from each other, which effectively means that there would be no repulsion to drive the motor. The coil is therefore connected to the circuit by contacts called commutators that allow the circuit to swap direction as each half of the coil spins past. This swaps the coil's polarity too, so it continually presents like (repelling) poles to the permanent magnet.

1827 William Prout classifies food into **carbohydrates, fats, and proteins**

1827 Robert Brown observes **Brownian motion**

1827–38 John James Audubon publishes *The Birds of America*

Friedrich Wöhler discovers **Wöhler synthesis** of urea

April 27 London Zoo opens as a scientific institution

Berzelius produces an **improved table** of atomic weights

Jön Jakob Berzelius **isolates the element thorium**

Ányos Jedlik creates the world's **first electric motor**

George Green outlines the first **mathematical theory of electricity and magnetism**

December Mary Anning finds fossil **pterosaur**